BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 030 · Half adders and full adders

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When we group four neighbouring 1s on a Karnaugh map, what happens to the Boolean expression?

Q2. Represent −13 using 8-bit sign and magnitude.

Q3. Insert 40, 20, 60, 10, 30, 50 into an empty binary search tree. Draw the resulting tree.

Q4. Using a 5-bit two’s complement mantissa and 4-bit two’s complement exponent, calculate A − B. Show exponent alignment and mantissa subtraction. A = $0.1100_2 \times 2^3$; B = $0.1000_2 \times 2^2$.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — half adder

Recall the term — full adder

Recall the term — Sum

Recall the term — Carry

Recall the term — carry-in

Recall the term — carry-out

Recall the term — XOR

New learning

Start with binary addition — One binary column can produce a sum bit and a carry bit.

Meet the half adder — A half adder adds two single-bit inputs: A and B.

Discover the two gates — XOR calculates Sum; AND calculates Carry.

Build its truth table — Test every possible pair of inputs.

Find the limitation — What happens when a previous column sends a carry into this column?

First half adder — Add A and B to produce an intermediate Sum S_1 and Carry C_1.

Second half adder — Feed S_1 and Carry-in into a second half adder. Its Sum is the final Sum.

Combine both carries — Feed C_1 and C_2 into an OR gate to obtain Carry-out.

Worked example

Model 1 + 1 with a half adder

1. Rightmost result bit — CODE:
Sum = 0

2. Next column — CODE:
Carry = 1

3. Trace XOR and AND — CODE:
1 XOR 1 = 0 → Sum
1 AND 1 = 1 → Carry

4. Connect both half adders — CODE:
S_1 = A XOR B

Sum = S_1 XOR Cin
Cout = (A AND B) OR (S_1 AND Cin)

We Do · Practise with guidance

Half-adder expressions

OCR asks for S and C.

Complete the same formulae

```
S = A ? B
C = A ? B
```

```
S = A XOR B
C = A AND B
```

Half-adder truth table

Complete the four outputs.

Fill the same table

```
A B | S C
0 0 | ? ?
0 1 | ? ?
1 0 | ? ?
1 1 | ? ?
```

```
A B | S C
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1
```

From half to full adder

Trace the two XOR gates, then the two AND gates feeding OR.

Improve the same explanation

```
Two half adders plus an ? gate can form a full adder.
```

```
Two half adders plus an OR gate can form a full adder.
```

Full-adder case

A=1, B=1, CarryIn=1.

Complete the output

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Half-adder question
State the Sum/Carry expressions, complete the truth table, then explain how half adders contribute to a full adder.

Question 2

Complete eight cases
Without looking at the diagram, sketch the five gates and their connections from memory. Then create the full-adder truth table for A, B and CarryIn.

Question 3

Explain four-bit addition
Explain how four full adders can add two 4-bit binary numbers.

Question 4

Add 0111 + 0001
Use bit-by-bit addition and identify where the carry travels.

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

For a half adder, Sum is produced by XOR and Carry is produced by AND.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The Boolean expression becomes simpler. Variables that change within the group can be removed, leaving only variables that stay the same.

Q2. Step 1: 13 in binary is 1101. Step 2: pad the magnitude to seven bits: 0001101. Step 3: because the number is negative, put 1 in the sign bit. Answer: 1 | 0001101 = 10001101. This is sign and magnitude , not two’s complement. SIGN · 1 BIT 1 Negative MAGNITUDE · 7 BITS 0001101 13 in binary

Q3. Insert the values in the order given . 40 becomes the root. 20 goes left of 40; 60 goes right of 40. 10 goes left of 20; 30 goes right of 20; 50 goes left of 60. Every smaller value goes to the left and every larger value to the right. 40 20 60 10 30 50

Q4. Align exponents: $B = 0.0100_2 \times 2^3$. Subtract mantissas: $0.1100_2 - 0.0100_2 = 0.1000_2$. Answer: $0.1000_2 \times 2^3 = 4$.

You Do pages · Independent questions

Q1. Half-adder question
CODE:
S = A XOR B
C = A AND B

A B | S C
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1

Two half adders can be joined, with an OR gate combining carry outputs, to form a full adder.
Check: Use XOR for Sum, AND for Carry and mention the OR connection.

Q2. Complete eight cases
CODE:
A B Cin | S Cout
0 0 0 | 0 0

0 0 1 | 1 0
0 1 0 | 1 0
0 1 1 | 0 1
1 0 0 | 1 0
1 0 1 | 0 1
1 1 0 | 0 1
1 1 1 | 1 1
Check: The outputs match ordinary binary addition of three bits.

Q3. Explain four-bit addition
CODE:
Use one full adder for each bit position. The carry-out from a lower-order full adder is connected to the carry-in of the next higher-order full adder. This propagates carries across the four-bit addition.
Check: Your answer connects carry-out from one bit to carry-in of the next.

Q4. Add 0111 + 0001
CODE:
0111
+ 0001

1000

A carry is generated at the least-significant column and propagates through the next two columns.
Check: The final result is 1000.

Exit Ticket · Model answer

True. A half adder uses XOR for Sum and AND for Carry.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding